Md. Minhazul Haq

Graduate Research Assistant, Universiti Malaya, Malaysia Portfolio Google Scholar GitHub
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Research Interests

Embedded AI, Computer Vision, Deep Learning, Biomedical Signal Processing and Autonomous Systems

Publications

1. Md. Minhazul Haq, Anis Salwa Mohd Khairuddin, Effariza Hanafi, Hafiz Muhammad Fahad Noman, and Mohamad Haniff Junos. *HAT-YOLO: Hybrid Attention and Transformer-Integrated Model for UAV Object Detection on Embedded Devices.* KSII Transactions on Internet and Information Systems, 20(2):920–941, 2026.
2. Yang Z, Khairuddin AS, Huang CJ, Ru WW, Noman HM, Putri TW, Xu X, Haq MM. *A Review of Short-Term, Long-Term, and Memory Consolidation Mechanisms in the Hippocampus and Cerebral Cortex.* IEEE Access, 2025.
3. Md. Minhazul Haq, Anis Salwa Mohd Khairuddin, Effariza Hanafi. *EdgeLite-YOLOv8n: An Efficient YOLOv8n Model for Real-Time Vehicle Detection in UAV Imagery on Embedded Platforms.* Accepted at the 2nd International Transdisciplinary Research Conference 2026, Universiti Malaya.
4. Md Minhazul Haq, Anis Salwa Mohd Khairuddin, Effariza Hanafi. An Optimized YOLOv8n Model for Real-Time Small-Object Detection on Resource-Constrained Platforms. Review in progress (Multimedia Tools And Applications Journal), Submission Id:- MTAP-D-26-02236.
5. Md Minhazul Haq, Anis Salwa Mohd Khairuddin, Effariza Hanafi. Towards Real-Time Wood Defect Detection on Resource-Constrained Devices: An Optimized YOLOv8n Model. Review in progress (Forest Science), Submission Id:- FRSC-D-26-00118.
6. Hasan, M. M., K, S., A, M., & Haq, M. M. *Development of Prototype Dual-Axis Solar Tracking Device for Optimal Power Generation at Residential Area in Bangladesh.* Journal of Energy and Safety Technology, 7(1):25–33, 2024.

Poster Presentation: Md. Omar Faruk, Md. Minhazul Haq, Anis Salwa B. M. Khairuddin, Effariza Hanafi, Karen L. Pua, and W. M. Tilakaratne. *Real-Time Oral Cancer Detection Using Lightweight YOLOv8n-Based Deep Learning and Computer Vision: A Pilot Study.* UMHEALTH Academic Month & Research Carnival 2025, Universiti Malaya.

Research Experience

Graduate Research Assistant Nov. 2023 – Feb. 2026
Centre of Intelligent Systems for Emerging Technology (CISET), Universiti Malaya

- Conducted research on embedded AI, computer vision, and autonomous perception, developing lightweight deep learning models for real-time object detection, monocular distance estimation, and edge-device deployment.
- Developed deep learning models for automated stress detection using photoplethysmography (PPG) signals.
- Evaluated and optimized state-of-the-art deep learning models for computer vision and biomedical AI, improving accuracy, efficiency, and real-time performance on resource-constrained devices.

Education

Master of Engineering Science (Research) Nov. 2023 – July 2026
Department of Electrical Engineering, Universiti Malaya, Malaysia
Field of Research: Artificial Intelligence
Thesis: *Optimization of Lightweight YOLO Model for Multiclass Object Detection on Embedded Device*

B.Sc. in Electrical and Electronics Engineering Final Year GPA: 3.55/4.00 | CGPA: 3.11/4.00
Jashore University of Science and Technology (JUST), Bangladesh Graduated: Apr. 2022
Thesis: *Design of Solar Tracking System for Optimal Power Generation at Residential Area in Bangladesh*

Teaching Experience

Graduate Teaching Assistant / Lab Demonstrator Oct. 2025 – Feb. 2026
LAB 2 (KIE1002), Department of Electrical Engineering, Universiti Malaya

Technical Skills

Programming: Python, C/C++, MATLAB, HTML
AI/ML: YOLO, RF-DETR, CNN, RNN, Computer Vision
Tools: OpenCV, PyTorch, Docker, Embedded Systems (e.g., Jetson Nano)

Awards

- **Best Presenter (Postgraduate Oral Presentation) – Winner**, 4th International Orthopaedic Postgraduate Symposium (IOPS4), Universiti Malaya, 29 Oct. 2025.
- **Best Project Award** – Voice-Controlled Humanoid Robot, Science Fair 2019, Jashore University of Science and Technology, Bangladesh.
- **Best Project Award** – Solar Scooter, Student to Startup 2.0 Campus Hunt, JUST, Bangladesh, Oct. 2019.

Standardized Test Scores

IELTS – April 8, 2025: Overall 7.0 (Speaking 7.5, Listening 7.5, Reading 7.0, Writing 6.0)